

Saturated Superfluid Vacuum IV

Gravity as Update-Capacity Gradient

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Abstract

This paper develops gravity in the Saturated Superfluid Vacuum (SSV) framework as a mechanical consequence of gradients in local update capacity. The central idea is that motion and interaction are not independent processes imposed on an inert background, but competing uses of the same local dynamical bandwidth of the vacuum medium. Where matter loads the medium with internal interaction, less update capacity remains for free propagation. Spatial gradients in this remaining capacity bend trajectories and appear as gravitational attraction.

The aim of the paper is to state this mechanism clearly and narrowly. We focus on the relation between local change-rate, time-dilation gradients, pressure gradients, and path bending. We do not attempt here a full black-hole ontology, cosmology, or galactic phenomenology.

1 Introduction

This paper develops gravity in SSV as a mechanical consequence of local update-capacity gradients. The main question is not how to quantize a pre-existing gravitational field, but how gravitational behavior arises if the vacuum is a real medium whose local rate of change is finite and load-dependent.

2 Local Change-Rate and Dynamical Bandwidth

Introduce the idea that proper time corresponds to local change-rate and that internal interaction and translational propagation are two uses of the same local dynamical capacity.

3 Gravity as Refraction Through Loaded Vacuum

Develop the core mechanical picture: matter loads the medium, reduces available propagation capacity nearby, and creates a gradient through which wavefronts and trajectories bend.

4 Time-Dilation Gradients

Connect the gravity mechanism to time-dilation language and clarify how reduced update capacity appears as slower local change.

5 Consequences

Discuss the intended scope of the mechanism for:

- free fall,
- light bending,
- gravitational redshift.

6 Discussion and Open Problems

Clarify what is derived, what is heuristic, and what still needs formal development.

7 Conclusion

Gravity in SSV is treated as the bending of motion through gradients in the local update capacity of the vacuum medium.